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Markets for Short-Circuit Current Services

Peng Wang

Technical University of Madrid

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What is SCC and why is its decline a problem ?

02 DR Makes SCC worse

Reshaping Load profiles may compromise SCC

03 How to Price SCC Services

How to incentivize generators to provide SCC ?

04 Key Takeaways & Future Work



The SCC Challenge

Definition of SCC

Short-Circuit Current (SCC) is the electrical current that flows during a short-circuit fault (e.g., line-to-ground fault) in a power system.

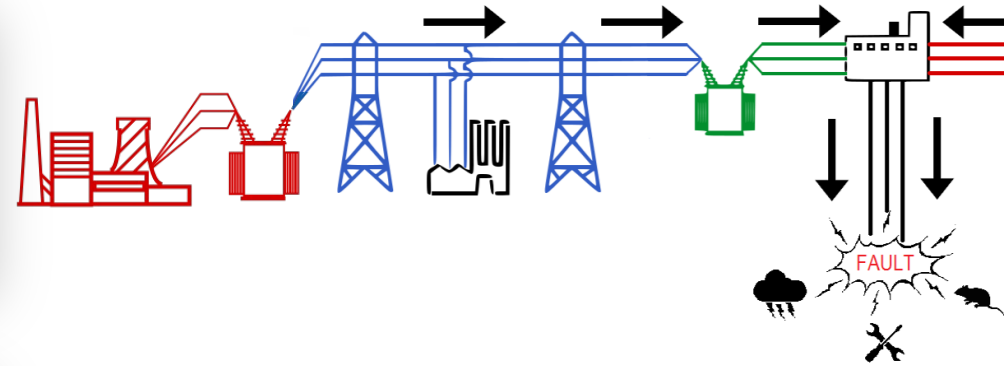
Why It Matters

- **Protective Relay Operation**

SCC is essential for relays to **"see" and detect faults**, triggering circuit breakers to isolate issues. **Low SCC** cause faults to go undetected, potentially cascading **blackouts**.

- **System Stability**

Adequate SCC maintains voltage stability during faults and prevents Inverter-Based Resources (IBR) from losing synchronization with the grid.



The SCC Challenge

The Modern Challenge

- Wind and solar IBR are **replacing** traditional Synchronous Generators (SGs) in modern power systems.
- Critical Gap: IBR contribute about **1/5** of the SCC of an equivalent SG.
- As SGs retire, the overall system SCC level is consequently **declining**, causing the protection relays to fail.

Synchronous machines

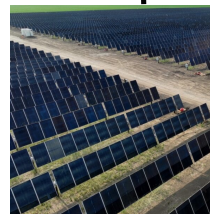


Magnetic energy

5–8 p.u. SCC

A **strong** electrical signal that can be easily detected by grid protective relays, ensuring they **trip reliably** and **isolate the fault quickly**.

Grid-following IBR



Electronic converters

1–3 p.u. SCC, even lower

A **weaker** fault current due to their electronic limits. This weak signal may be insufficient to trigger the protection, **leaving faults unchecked and endangering grid stability**.

For a power system incorporating SGs and IBR, the SCC at bus b due to their current injections can be expressed as:

$$I_{b\text{sc}} = \frac{\sum_{g \in G} (Z_{b\Psi(g)} I_g u_g) + \sum_{c \in C} (Z_{b\Phi(c)} I_c \alpha_c)}{Z_{bb}}$$

Chu, Zhongda, and Fei Teng. "Short circuit current constrained UC in high IBG-penetrated power systems." *IEEE Transactions on Power Systems* 36.4 (2021): 3776-3785.

SCC is being decreasing

Explicit SCC constraints are required ! >>

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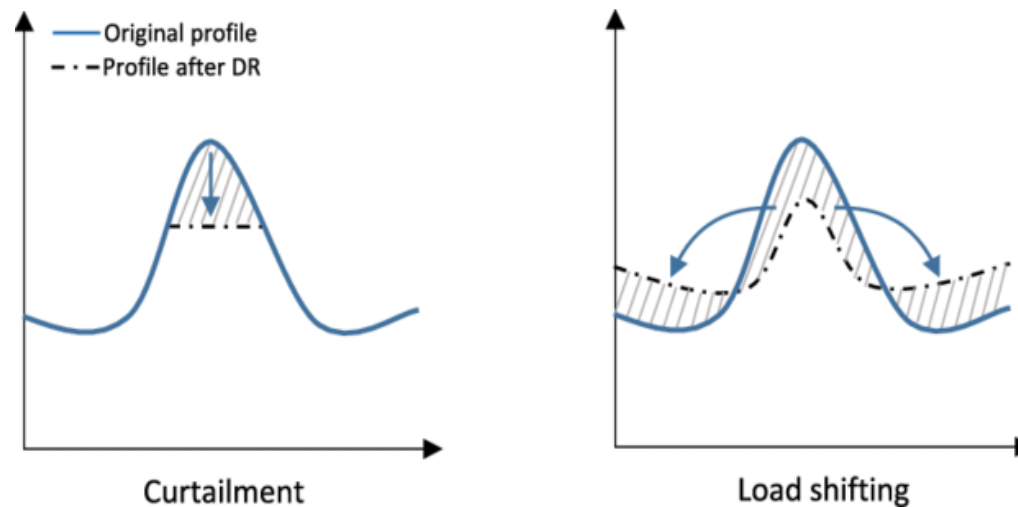
03 How to Price SCC Services

How to incentivize generators to provide SCC ?



DR Makes SCC Worse

Demand Response (DR) is economically beneficial to grids



Wang, Peng, and Luis Badesa. "Pricing Short-Circuit Current via a Primal-Dual Formulation for Preserving Integrality Constraints." PESGM 2026, arXiv preprint arXiv:2510.05293 (2025).

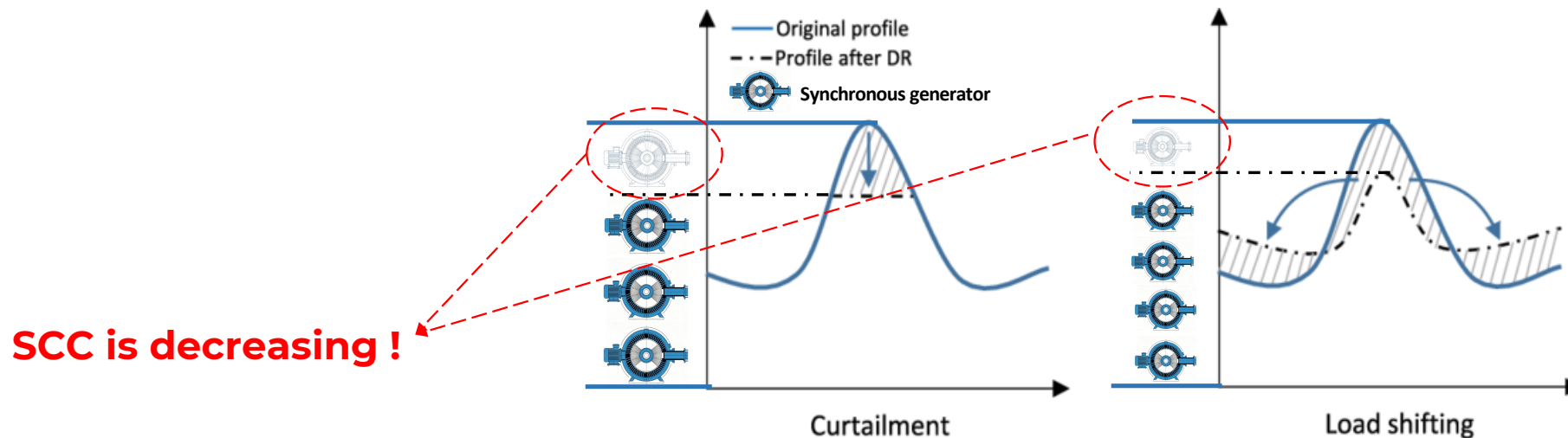


DR Makes SCC Worse

The "Double-Edged Sword" of DR

Demand Response (DR) is economically beneficial to grids, but may reduce system load, leading to fewer SGs being online.

This exacerbates the SCC deficiency.



Wang, Peng, and Luis Badesa. "Pricing Short-Circuit Current via a Primal-Dual Formulation for Preserving Integrality Constraints." PESGM 2026, arXiv preprint arXiv:2510.05293 (2025).

DR Makes SCC Worse

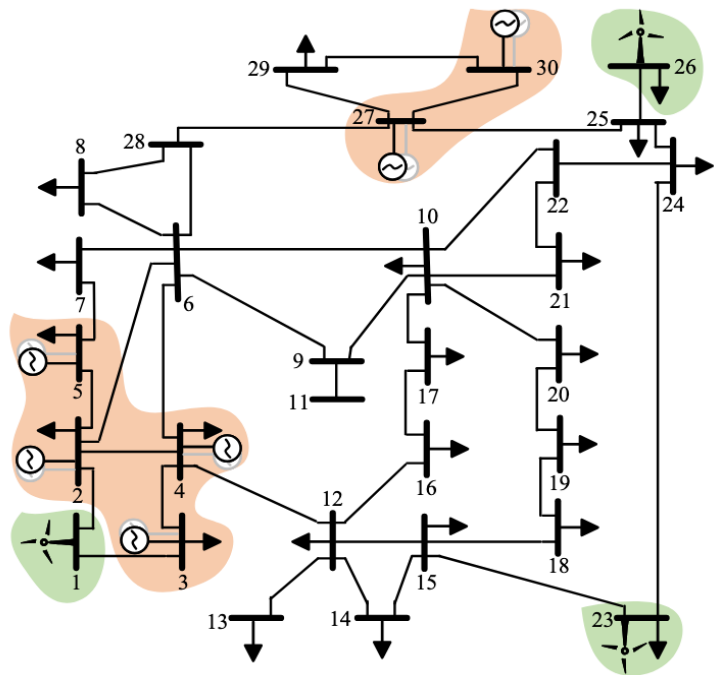


Fig. Modified IEEE 30-bus system



SCC constraints

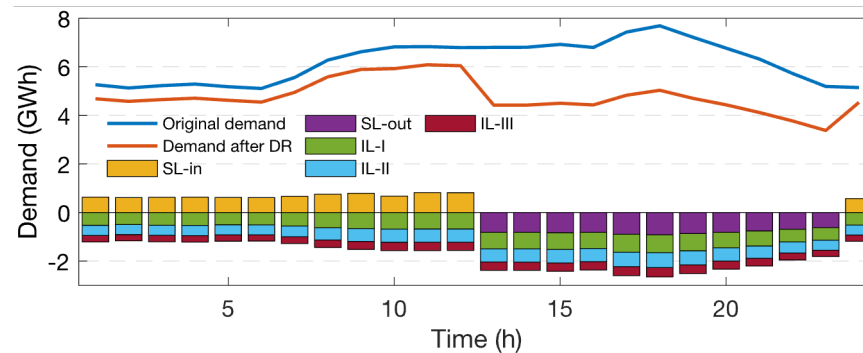


Fig. System demand in the case of no SCC constraints.

- Interruptible load is **curtailed** for compensation to lower electricity costs.
- Shiftable load **moves** from peak hours to off-peak to reduce bills.

Wang, Peng, and Luis Badesa. "Pricing Short-Circuit Current via a Primal-Dual Formulation for Preserving Integrality Constraints." PESGM 2026, arXiv preprint arXiv:2510.05293 (2025).



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DR Makes SCC Worse

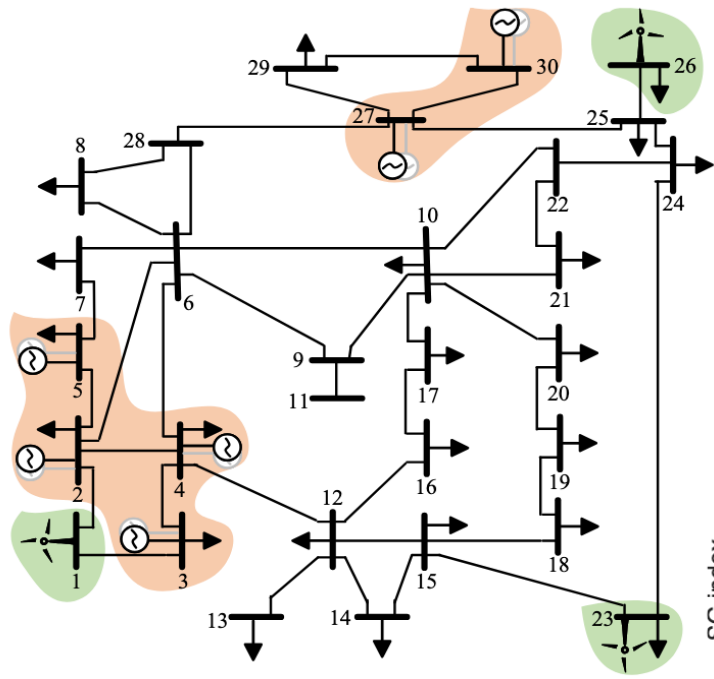


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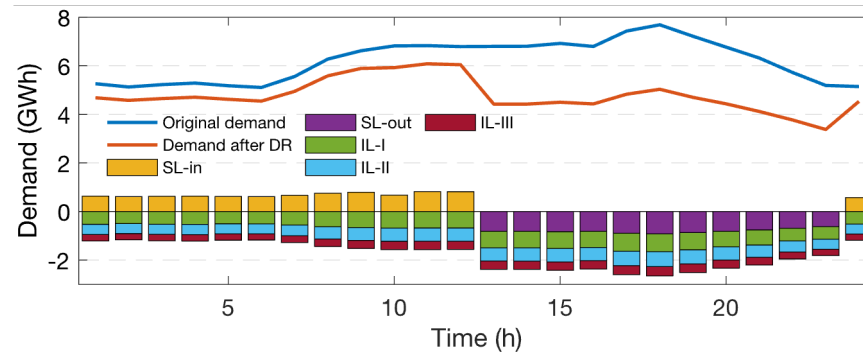


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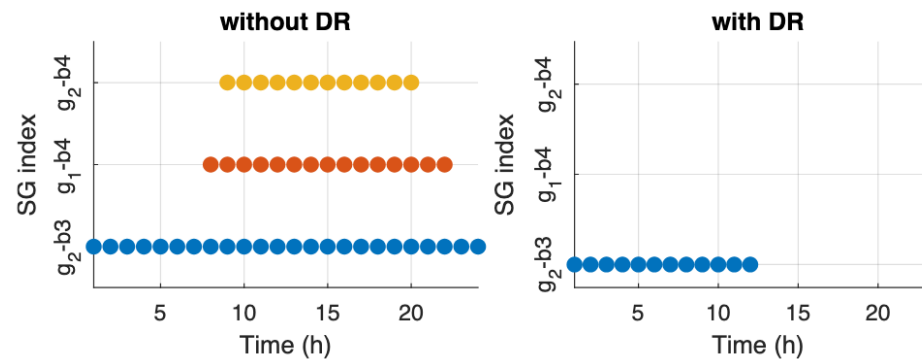


Fig. Generators whose commitment state is affected by DR.

The reduced load **lowers generator dispatch**: one (●) operates at reduced output, while two (●●) are shut down throughout the day.

Wang, Peng, and Luis Badesa. "Pricing Short-Circuit Current via a Primal-Dual Formulation for Preserving Integrality Constraints." PESGM 2026, arXiv preprint arXiv:2510.05293 (2025).

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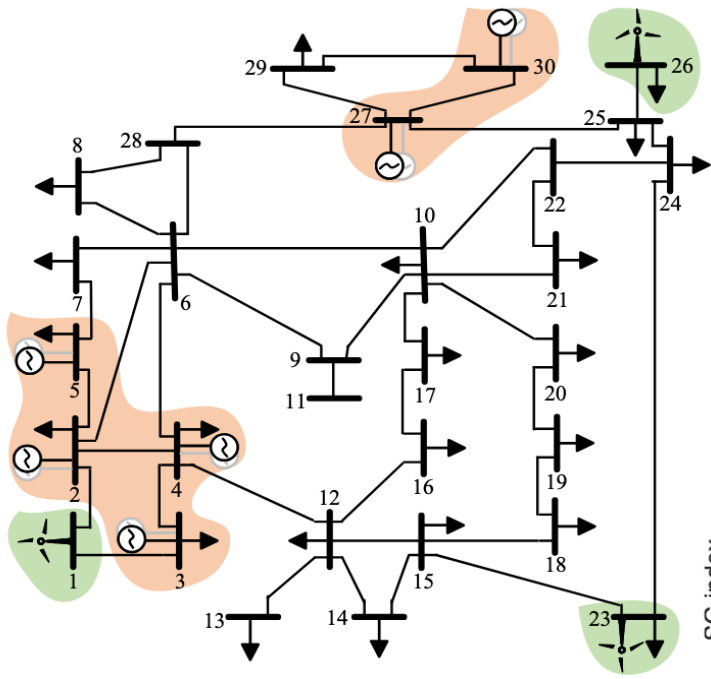


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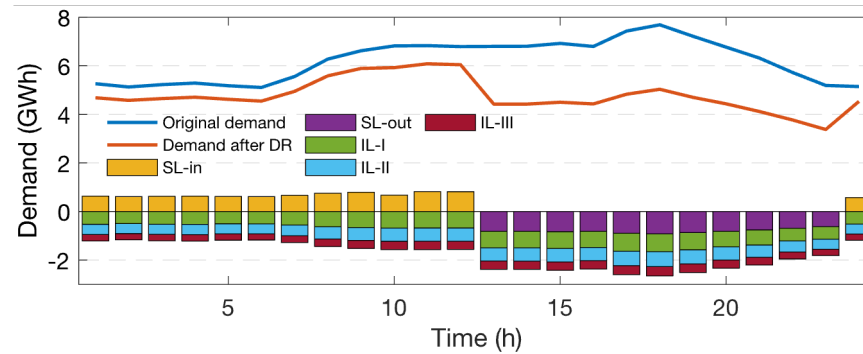


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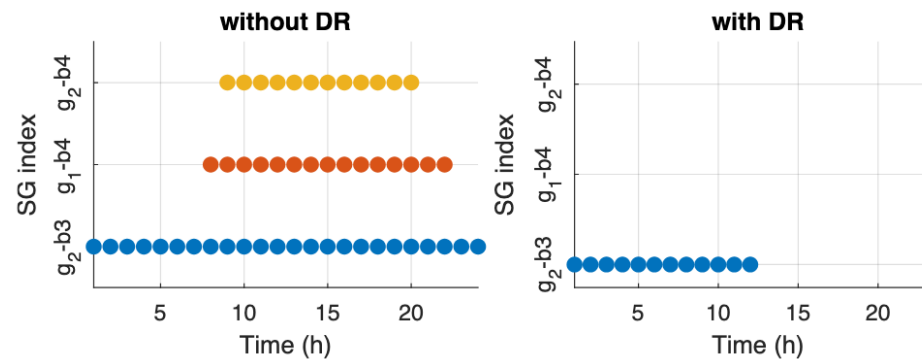


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How about SCC of the system ? >>

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DR Makes SCC Worse

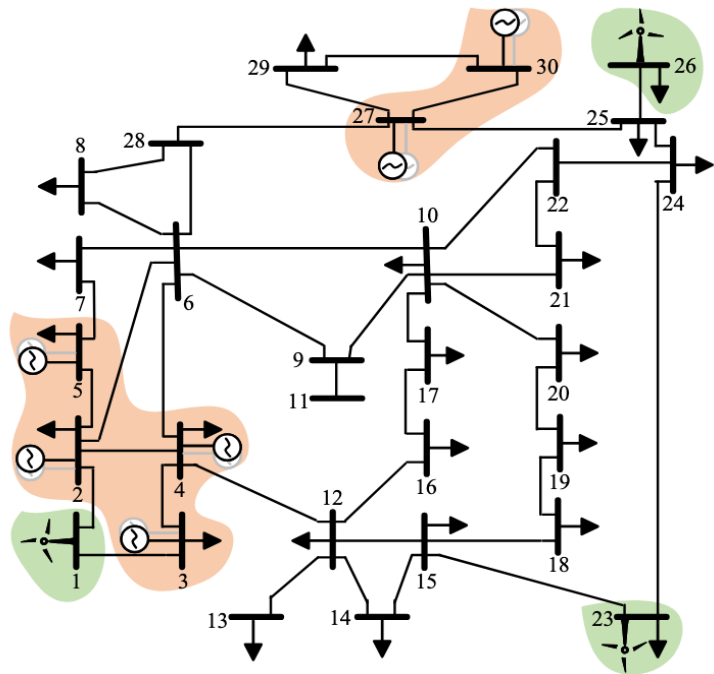


Fig. Modified IEEE 30-bus system

✗
SCC constraints

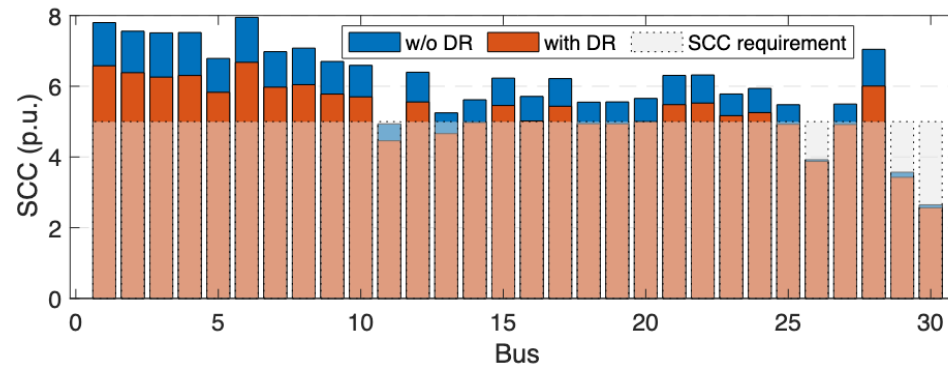


Fig. Unconstrained minimum SCC level at each bus.

Table. Buses with Inadequate SCC Level

DR implementation	Bus index
without DR	11, 26, 29, 30
with DR	11, 13, 14, 18, 19, 20, 25, 26, 27, 29, 30

SCC security level
for buses: 5 p.u.

Wang, Peng, and Luis Badesa. "Pricing Short-Circuit Current via a Primal-Dual Formulation for Preserving Integrality Constraints." PESGM 2026, arXiv preprint arXiv:2510.05293 (2025).

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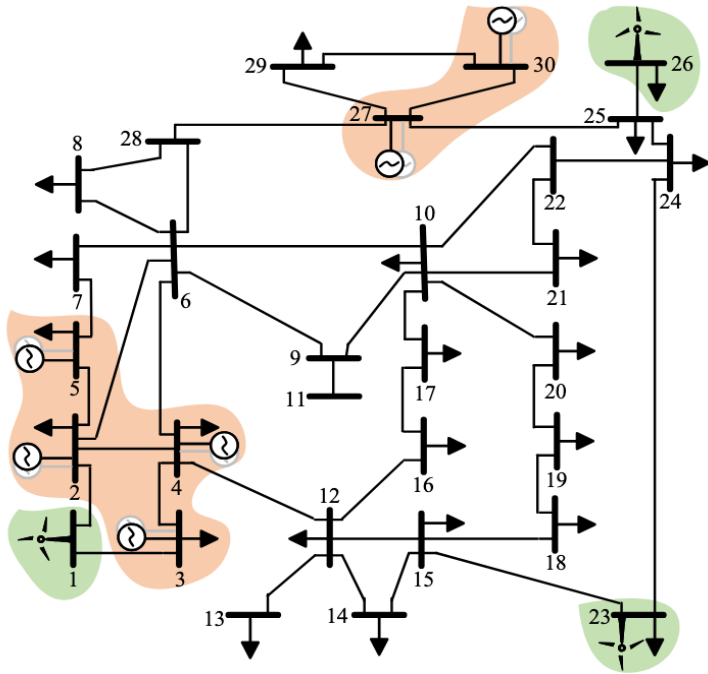


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SCC constraints

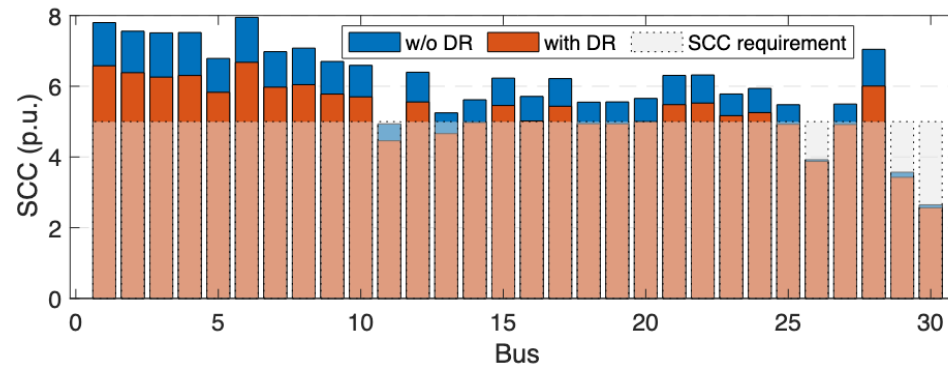


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Table. Buses with Inadequate SCC Level

DR implementation	Bus index
without DR	11, 26, 29, 30
with DR	11, 13, 14, 18, 19, 20, 25, 26, 27, 29, 30

DR leads to insufficient SCC in more buses.

Explicit SCC constraints are required ! >>

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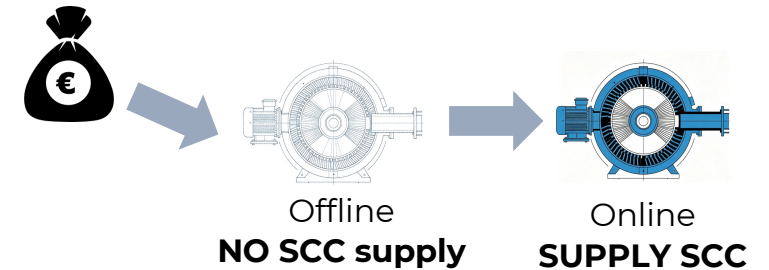
How to incentivize generators to provide SCC ?

04 Key Takeaways & Future Work



How to Price SCC Services

Pricing SCC is relevant, as it places necessary **financial incentives** to generators for SCC supply when needed.



Wang, Peng, and Luis Badesa. "Pricing Short-Circuit Current via a Primal-Dual Formulation for Preserving Integrality Constraints." *arXiv preprint arXiv:2510.05293* (2025).



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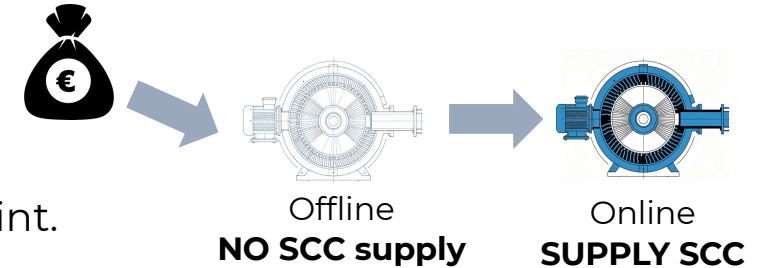
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How to Price SCC Services

Pricing SCC is relevant, as it places necessary **financial incentives** to generators for SCC supply when needed.

Pricing SCC is basically computing dual variable of SCC constraint.

$$I_{b_{SC}} = \frac{\sum_{g \in \mathcal{G}} Z_{b\Psi(g)} I_g u_g + \sum_{c \in \mathcal{C}} Z_{b\Phi(c)} I_c \alpha_c}{Z_{bb}}$$

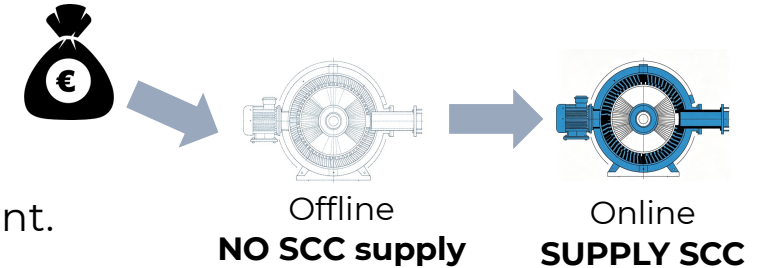


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Offline training to approximate actual SCC

$$I_{b_L} = \sum_g k_{bg} u_g + \sum_c k_{bc} \alpha_c + \sum_m k_{bm} \eta_m \geq I_{b_{lim}}$$

$$\eta_m = u_{g_1} \cdot u_{g_2}, \quad \text{s.t. } \{g_1, g_2\} = m$$

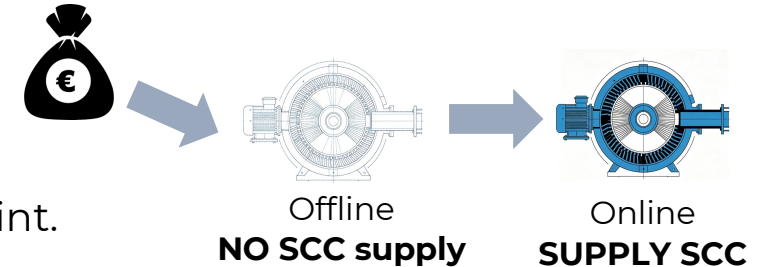
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Inherent **discrete** (binary) variables **prevent price formation**.

Because the **prices** derived from duality theory must need the problem to be **continuous and convex**.

Duality in optimization, KKT and shadow prices – With applications to electricity markets (<https://doi.org/10.5281/zenodo.16504513>)

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How to address binary variables ? >>

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How to Price SCC Services

Primal-Dual Formulation:

A formulation that tries to **minimize the duality gap**, considering non-negative profit for each unit:

Primal of SCC-Constrained UC
(Non-convexity from binary variables)

1

1: Decide **quantity** of products, i.e., energy and SCC

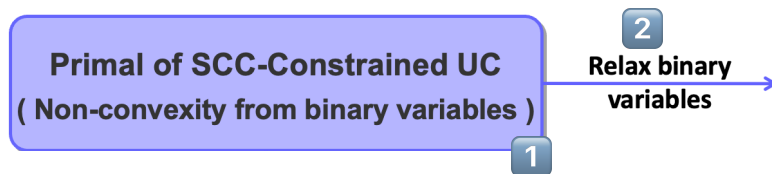
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How to Price SCC Services

Primal-Dual Formulation:

A formulation that tries to **minimize the duality gap**, considering non-negative profit for each unit:



- 1: Decide **quantity** of products, i.e., energy and SCC
- 2: Make the whole problem convex and continuous

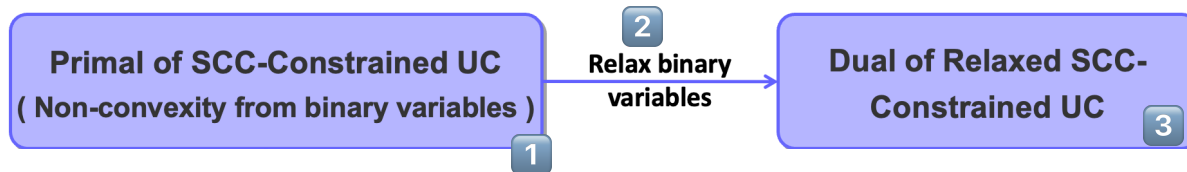
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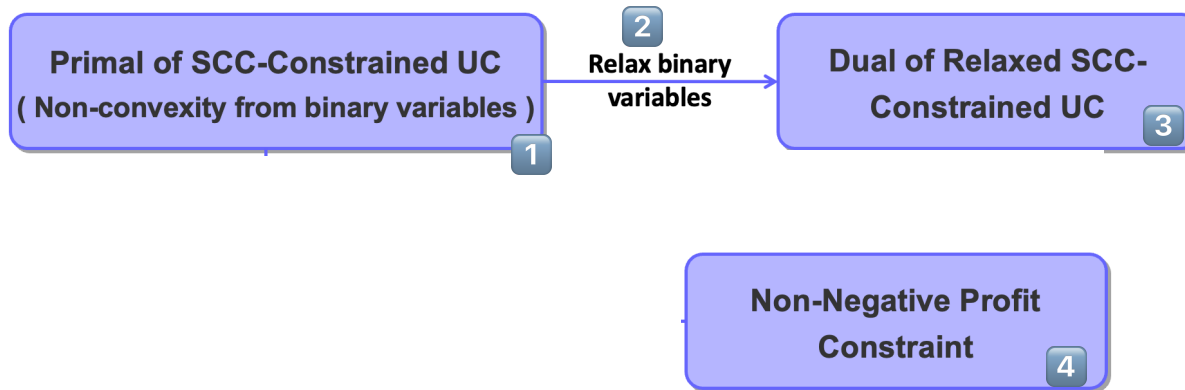
- 1: Decide **quantity** of products, i.e., energy and SCC
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- 3: Decide **prices** of products, i.e., energy and SCC

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$$\underbrace{\lambda^E P_g}_{\text{Energy revenue}} + \underbrace{\sum_b \lambda_b^{\text{SCC}} k_{bg} u_g + \sum_b \sum_{\{m|g \in m\}} \lambda_b^{\text{SCC}} k_{bm} \eta_m}_{\text{SCC revenue}} - \underbrace{(c_g^{\text{nl}} u_g + c_g^{\text{m}} P_g + C_g^{\text{rst}})}_{\text{Operating cost}} \geq 0, \quad \forall g$$

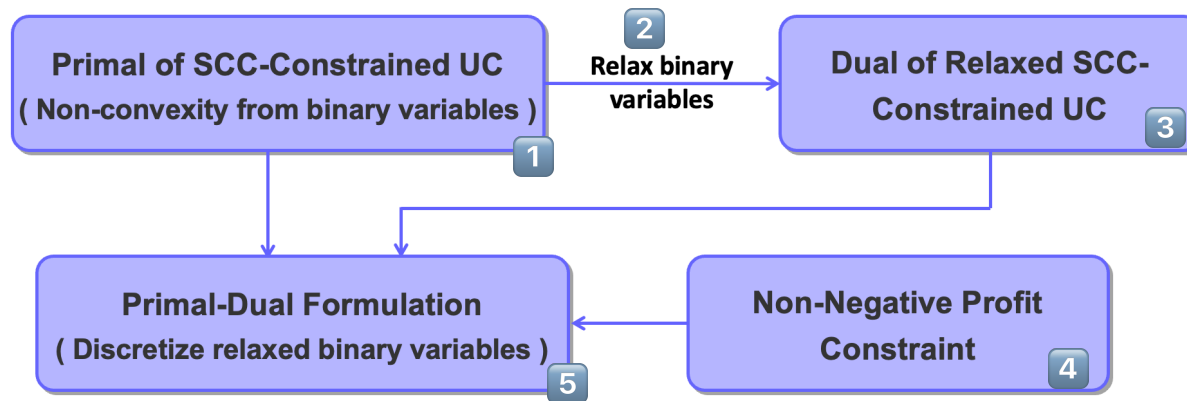
- 1: Decide **quantity** of products, i.e., energy and SCC
- 2: Make the whole problem convex and continuous
- 3: Decide **prices** of products, i.e., energy and SCC
- 4: Make sure the prices are **revenue-adequate**

Wang, Peng, and Luis Badesa. "Pricing Short-Circuit Current via a Primal-Dual Formulation for Preserving Integrality Constraints." *arXiv preprint arXiv:2510.05293* (2025).

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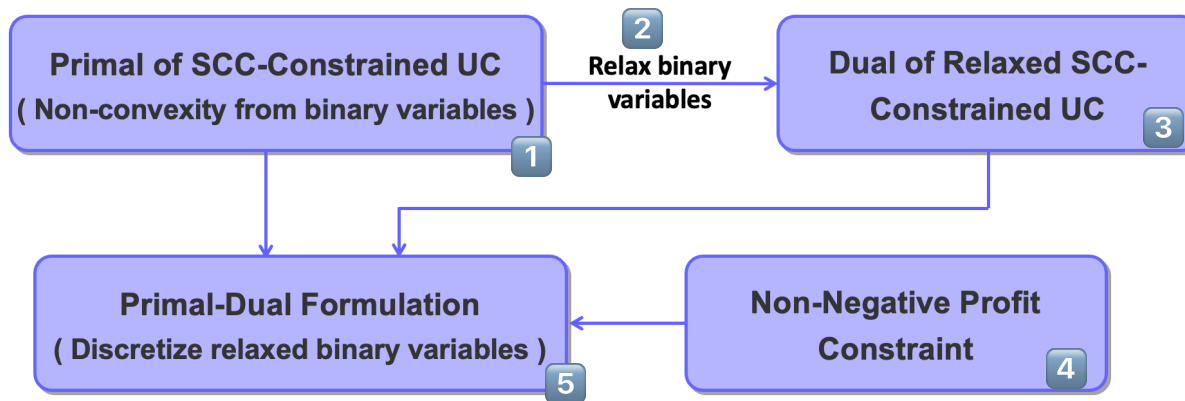
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- 5: Coordinate the primal and dual problems to **co-optimize quantity and price**

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$$\min_V \text{Obj}_{\text{primal}} - \text{Obj}_{\text{dual}}$$

where:

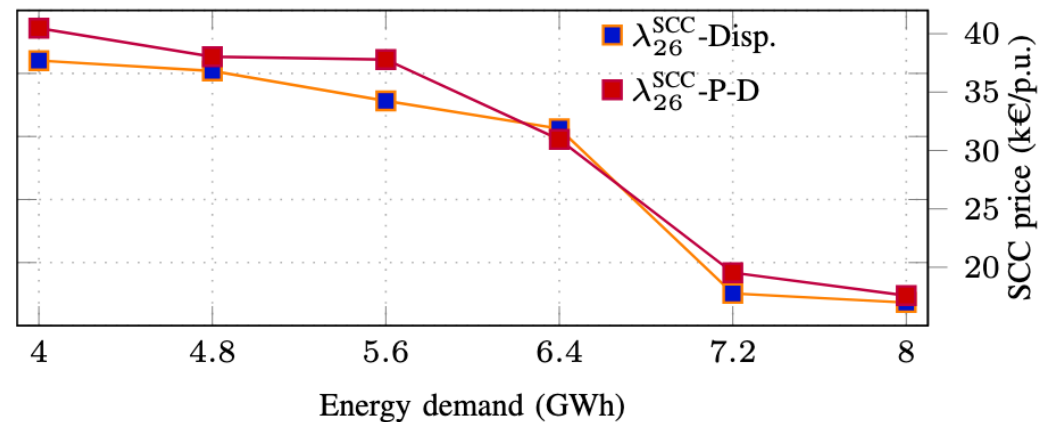
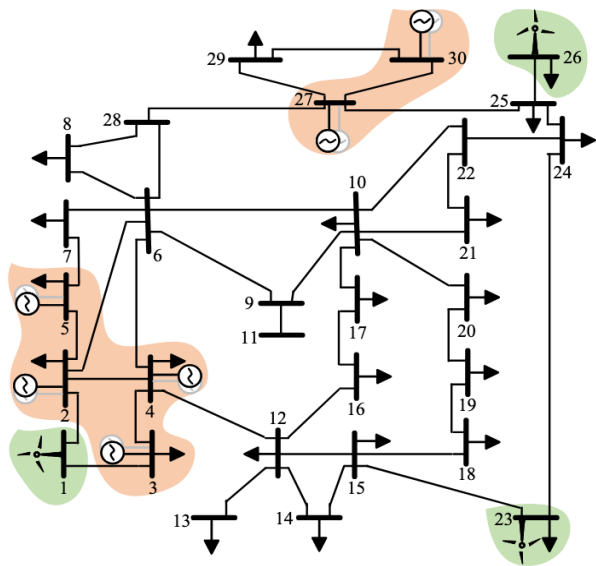
$$V = \{ V_{\text{primal}}, V_{\text{dual}} \}$$

subject to:

- Primal constraints
- Dual constraints
- Non-negative profit constraint

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How to Price SCC Services



Dispatchable pricing is a method which simply relaxes binary variables to compute prices

The SCC price derived from P-D is higher than the dispatchable

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How to Price SCC Services

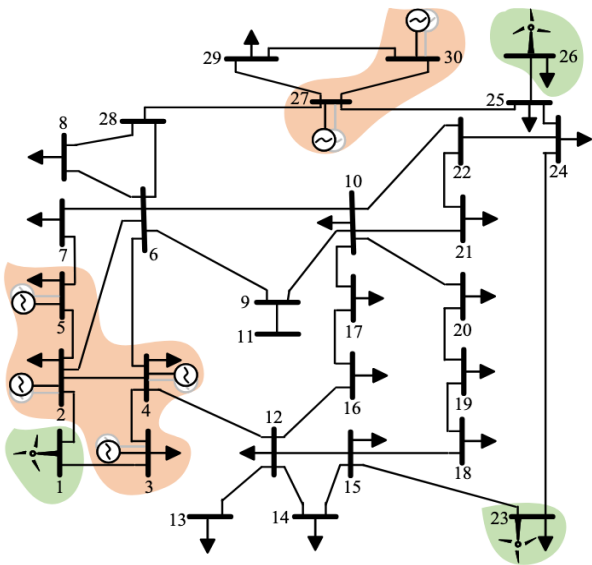


Table. Revenues and profits of SGs under *dispatchable pricing method* (k€)

Bus of SG	Energy profit	SCC revenue	Profit
Bus 2	-1.74	1.73	-0.01
Bus 4	-2.40	2.23	-0.17
Bus 30	-4.43	4.37	-0.06

Table. Revenues and profits of SGs under *primal-dual formulation*

Bus of SG	Energy profit	SCC revenue	Profit
Bus 2	-1.72	1.85	0.13
Bus 4	-2.39	2.39	0
Bus 30	-4.43	4.69	0.26

Primal-dual formulation decides prices that fully recover SGs' operating costs

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Key Takeaways & Future Work



Key Takeaways & Future Work

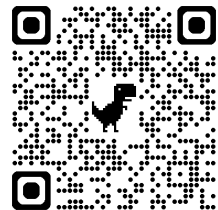
- *Sufficient SCC is essential for protective relays to detect short-circuit faults, yet high IBR penetration reduces system-wide SCC.*
- *DR may shut down excess synchronous units and further lower SCC.*
- *Primal-dual formulation can appropriately price SCC as an ancillary service to incentivize SCC provision.*

Key Takeaways & Future Work

- *Sufficient SCC is essential for protective relays to detect short-circuit faults, yet high IBR penetration reduces system-wide SCC.*
- *DR may shut down excess synchronous units and further lower SCC.*
- *Primal-dual formulation can appropriately price SCC as an ancillary service to incentivize SCC provision.*

What is next

- *How to price other stability services such as voltage stability ?*
- *Markets generally operate under imperfect competition. How do strategic market agents affect system and market operations, particularly through **market power** ?*



Wang Peng, and Luis Badesa. "Imperfect Competition in Markets for Short-Circuit Current Services." arXiv preprint arXiv:2508.09425 (2025).

Acknowledgements



Luis Badesa

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